Component Diversity, Excess Entropy, and Divalent-Ion Self-Diffusion in Charged Lennard-Jones Mixtures

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Does increasing the number of mixture components raise excess entropy and accelerate divalent ions? We combine HNC screening of 35 candidates with replicated molecular dynamics of eight selected charged Lennard-Jones mixtures at $T^* = 2$, $\rho^* = 0.7$. Each mixture contains unchanged $+2/-2$ probe fractions of 0.03 per sign, with $N = 4000$ and three independent replicas. Excess entropy is obtained from 19-density thermodynamic integration with a fixed shifted LJ cutoff and a screened dilute endpoint; transport uses one common center-of-mass frame. High-entropy solvent designs raise MD excess entropy by approximately $0.30k_B$ per particle and divalent self-diffusion by 18–20% relative to the original baselines. A more controlled diversification, preserving charged composition and mean bare LJ pair amplitude, raises excess entropy by $0.103k_B$, mean self-diffusion by 8–9%, and probe diffusion by approximately 5%. The small ion effect is sensitive to MSD fitting window and unresolved with monovalent background. An exact label-only split changes formal ideal mixing entropy but leaves diffusion identical; adding monovalent salt alone gives no resolved probe enhancement. Both mean and probe diffusion show positive descriptive Rosenfeld-like trends, with weaker probe slopes. Residual charge-sign asymmetry and finite-size uncertainty limit quantitative claims. The results identify faster-ion, higher-excess-entropy mixtures, but establish neither a causal component-count effect nor a universal species scaling law. A corrected classical 2PT analysis passes internal and label checks but retains composition-dependent entropy biases, supporting spectral diagnostics while leaving thermodynamic integration as the primary entropy estimate.

I. INTRODUCTION

Excess-entropy scaling offers a link between equilibrium correlations and liquid transport. Rosenfeld proposed that an appropriately reduced diffusion coefficient obeys [5, 6]

$$D_i^R = A_i \exp(B_i s_{\rm ex}/k_B), \qquad D_i^R = D_i \rho^{1/3} \sqrt{m_i/(k_B T)}. \tag{1}$$

For mixtures, whether this relation describes individual charged species as well as the solvent-dominated mean is a separate question. Local solvation, association, and interaction-dependent friction can respond differently to composition. Generalized tracer scaling includes additional kinetic information and is not guaranteed when component mobilities decouple [7].

The entropy entering Eq. (1) is the excess per particle relative to the ideal mixture at the same temperature, density, and composition. The ideal reference already includes $s^{\rm id}_{\rm mix}/k_B = -\sum_i x_i \ln x_i$. A greater number of species can therefore increase total entropy without increasing $s_{\rm ex}$ or changing any force. Distinguishing label multiplicity from physical interaction diversity is essential to testing a component-induced speedup.

Our earlier four-mixture comparison identified an inconsistent LJ cutoff along the density-integration path and incompatible archived ion-transport estimates. The fixed-cutoff correction was material for absolute entropy but could not resolve species mobility. Here we replace the earlier transport evidence with a larger, independently replicated HNC-guided study, including an exact label-only null and physically diverse solvents at matched mean bare interaction amplitude. The original cutoff audit is retained in Appendix A; its four-mixture transport and 2PT decompositions are not used in the present scaling analysis.

We distinguish the particle-averaged self-diffusion $D_{\rm net} = \sum_i x_i D_i$ from mutual diffusion and conductivity. The divalent quantity is the average self-diffusion of the two probe signs, not motion of neutral ion pairs. The central tests are whether physical diversification raises MD excess entropy, whether the probes accelerate reproducibly, and whether their response is as strong as that of the mean fluid.

II. MODEL AND THEORETICAL FRAMEWORK

A. Interaction Potentials and Mixture Specifications

We consider a classical fluid mixture of p components characterized by number densities $\rho_i = N_i/V$ $(i = 1, \ldots, p)$, with total number density $\rho = \sum_{i=1}^{p} \rho_i$ and mole fractions $x_i = \rho_i/\rho$. The particles interact via a cut-and-shifted Lennard-Jones (LJ) core augmented by Coulomb interactions:

$$u_{ij}(r) = u_{ij}^{\rm LJ}(r) + \frac{q_i q_j e^2}{4\pi\varepsilon_0 \varepsilon_r r}, \tag{2}$$

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where the short-range LJ contribution is truncated and shifted at $r_c = 2.5\,\sigma$:

$$u_{ij}^{\mathrm{LJ}}(r) = \begin{cases} 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r}\right)^{12} - \left(\frac{\sigma_{ij}}{r}\right)^{6}\right] - u_{ij}^{\mathrm{shift}}, & r \le r_c, \\ 0, & r > r_c. \end{cases} \tag{3}$$

All particles share identical diameters $\sigma_{ij} = \sigma = 1.0$ and masses $m_i = m = 1.0$. Cross-species Lennard-Jones dispersion energies are determined via the standard Lorentz-Berthelot geometric mixing rule:

$$\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j}. \tag{4}$$

Reduced Lennard-Jones units are employed throughout: length $r^* = r/\sigma$, density $\rho^* = \rho\sigma^3$, temperature $T^* = k_B T/\varepsilon_1$, energy $u^* = u/\varepsilon_1$, and time $t^* = t/\tau$ with $\tau = \sigma\sqrt{m/\varepsilon_1}$.

The reduced Coulomb energy prefactor is $\Gamma = e^2/(4\pi\varepsilon_0\varepsilon_r\sigma\varepsilon_1) = 1.0$. The dimensionless Bjerrum length is therefore $\lambda_B^* = e^2/(4\pi\varepsilon_0\varepsilon_r\sigma k_B T) = \Gamma/T^* = 0.50$ at $T^* = 2.00$. The parameter Γ used in the input is an energy prefactor, not the Bjerrum length itself. The ionic strength of the mixture is given by

$$I = \frac{1}{2}\sum_{i=1}^{p} x_i q_i^2. \tag{5}$$

The Debye screening parameter κ is defined as

$$\kappa = \sqrt{4\pi\beta^*\rho^*\sum_{i=1}^{p} x_i q_i^2} = \sqrt{8\pi\beta^*\rho^* I}. \tag{6}$$

B. Ornstein-Zernike Equations and HNC Closure

The microscopic structure of the multicomponent fluid is described by the matrix of total correlation functions $h_{ij}(r) = g_{ij}(r) - 1$ and direct correlation functions $c_{ij}(r)$, coupled via the multicomponent Ornstein-Zernike (OZ) relations [1]:

$$h_{ij}(r) = c_{ij}(r) + \sum_{k=1}^{p}\rho_k \int c_{ik}(|\mathbf{r} - \mathbf{r}'|)h_{kj}(r')\,d\mathbf{r}'. \tag{7}$$

In Fourier space, defining $\tilde{h}_{ij}(k)$ and $\tilde{c}_{ij}(k)$ via standard three-dimensional Fourier transforms, Eq. (7) takes the matrix form

$$\tilde{\mathbf{H}}(k) = \tilde{\mathbf{C}}(k) + \tilde{\mathbf{C}}(k)\mathbf{P}\tilde{\mathbf{H}}(k), \tag{8}$$

where $\mathbf{P}_{ij} = \rho_i\delta_{ij}$ is the diagonal density matrix.

To treat the long-range Coulomb tail without numerical truncation artifacts, we employ the Allnatt screening decomposition [2, 3]. The pair potential is split into short-range (SR) and long-range (LR) parts:

$$u_{ij}(r) = u_{ij}^{\mathrm{SR}}(r) + u_{ij}^{\mathrm{LR}}(r), \tag{9}$$

with $u_{ij}^{\mathrm{LR}}(r) = q_i q_j \Gamma\psi(r)$, where $\psi(r) = \mathrm{erf}(\alpha r)/r$ and α is an optimized Ewald splitting parameter. The direct correlation function is decomposed accordingly:

$$c_{ij}(r) = c_{ij}^{\mathrm{SR}}(r) - \beta u_{ij}^{\mathrm{LR}}(r). \tag{10}$$

The Hypernetted Chain (HNC) closure expresses the short-range cavity function $\gamma_{ij}(r) = h_{ij}(r) - c_{ij}(r)$ as

$$g_{ij}(r) = \exp\left[-\beta u_{ij}(r) + \gamma_{ij}(r)\right], \tag{11}$$

which corresponds to setting the bridge function $B_{ij}(r) \equiv 0$. The resulting coupled non-linear integral equations are solved iteratively on a radial grid of $N_r = 4096$ points with grid spacing $\Delta r = 0.005\,\sigma$, using the multicomponent integral-equation solver [4].

C. Thermodynamic Properties and Free Energy Functional

Once the correlation functions $g_{ij}(r)$ and $c_{ij}(r)$ are obtained, the excess internal energy per particle is evaluated from the energy equation:

$$\frac{u_{\mathrm{ex}}}{k_B T} = \frac{1}{2}\beta\rho\sum_{i,j=1}^{p} x_i x_j \int_0^\infty 4\pi r^2 u_{ij}(r) g_{ij}(r)\,dr. \tag{12}$$

The compressibility factor $Z = P/(\rho k_B T)$ is obtained via the virial route:

$$Z_{\mathrm{virial}} = 1 - \frac{\beta\rho}{6}\sum_{i,j=1}^{p} x_i x_j \int_0^\infty 4\pi r^3 \frac{du_{ij}(r)}{dr} g_{ij}(r)\,dr. \tag{13}$$

Within the HNC approximation, the excess Helmholtz free energy per particle $a_{\mathrm{ex}} = A_{\mathrm{ex}}/(N k_B T)$ can be computed directly as a closed-form functional of the pair correlation functions without performing numerical parameter integrations [2, 10, 11]:

$$\begin{aligned} a_{\mathrm{ex}}^{\mathrm{HNC}} = &-\frac{1}{2\rho}\sum_{i,j=1}^{p}\rho_i\rho_j \int d\mathbf{r}\left[\frac{1}{2}h_{ij}^2(r) - c_{ij}(r)h_{ij}(r)\right. \\ &\left. + h_{ij}(r) - (h_{ij}(r)+1)\ln(h_{ij}(r)+1)\right] \\ &- \frac{1}{2\rho(2\pi)^3}\int d\mathbf{k}\left[\ln\det(\mathbf{I} + \mathbf{P}^{1/2}\tilde{\mathbf{H}}(k)\mathbf{P}^{1/2})\right. \\ &\left. - \mathrm{Tr}\left(\mathbf{P}\tilde{\mathbf{H}}(k)\right)\right]. \end{aligned} \tag{14}$$

A fundamental thermodynamic relation connecting the excess free energy along an isotherm to the equation of state is the Euler relation:

$$Z_{\mathrm{diff}} = 1 + \rho\left(\frac{\partial a_{\mathrm{ex}}}{\partial\rho}\right)_T. \tag{15}$$

Conversely, integrating the equation of state from the ideal gas limit along the isotherm yields the excess free energy via thermodynamic integration:

$$a_{\rm ex}(\rho) = \int_0^\rho \frac{Z(\rho') - 1}{\rho'}\, d\rho'. \tag{16}$$

Numerical differentiation and virial-route agreement are internal HNC consistency tests, as discussed in the accompanying theory report. They do not establish agreement with MD, nor do they supply transport coefficients. The new HNC screening values and replicated MD comparisons are reported in Section IV.

The excess entropy per particle $s_{\rm ex}/k_B$ then follows from the thermodynamic definition:

$$\frac{s_{\rm ex}}{k_B} = \frac{u_{\rm ex}}{k_B T} - a_{\rm ex}. \tag{17}$$

D. Green-Wallace Pair Structural Entropy

In addition to the total thermodynamic excess entropy $s_{\rm ex}$, we compute the two-body structural entropy $s^{(2)}$ from the Green-Wallace expansion [8, 9]. For a multicomponent mixture, $s^{(2)}/k_B$ is expressed as

$$\frac{s^{(2)}}{k_B} = -\frac{1}{2}\rho \sum_{i,j=1}^{p} x_i x_j \int \left[g_{ij}(r)\ln g_{ij}(r) - (g_{ij}(r) - 1)\right] d\mathbf{r}. \tag{18}$$

The integrand in Eq. (18) vanishes identically as $g_{ij}(r) \to 1$ at large distances, ensuring rapid convergence. As shown by Wallace [9], $s^{(2)}$ represents the loss of translational entropy associated with static spatial pair correlations.

III. HNC-GUIDED REPLICATED MOLECULAR DYNAMICS STUDY

A. Composition Design and Controls

The new study tests whether adding physically distinct components can raise excess entropy and accelerate unchanged divalent probes. All selected mixtures have $x_{+2} = x_{-2} = 0.03$, equal masses and nominal diameters, $T^* = 2$, and target density $\rho^* = 0.7$. Table I defines the eight states. B3 and B5 reproduce the compositions of the original three-component and monovalent-background five-component baselines. The new solvent mixtures contain 30% secondary neutral material, rather than the 5% used in the earlier four-component dispersion comparison; they are distinct physical states.

Thirty-five integer-charge, mole-fraction, and dispersion candidates were screened with HNC on 2048 radial points with $\Delta r = 0.01\sigma$. All converged. The eight selected candidates were refined on 4096 points with $\Delta r = 0.005\sigma$. Their excess entropies agree with the coarse results at the reported $10^{-4} k_B$ precision. The first 20 sampled low-wavevector structure-factor matrices were positive for these states; this is a screening diagnostic, not a proof of phase stability. Charge-only candidates did not provide the desired large positive excess-entropy signal, so the replicated MD effort concentrates on solvent diversification and the original salt-addition comparison.

U4 contains one secondary neutral solvent with $\varepsilon = 0.25$. L5 splits that solvent into equal populations of mechanically identical labels. M5 and H5 instead split it into two physically distinct solvents with respective ε pairs $(0.0625, 0.5625)$ and $(0.01, 0.81)$. Under geometric mixing, all four states have

$$\sum_i x_i \sqrt{\varepsilon_i} = 0.85, \qquad \sum_{ij} x_i x_j \varepsilon_{ij} = 0.85^2 = 0.7225. \tag{19}$$

Their ion fraction, ionic strength, nominal sizes, and masses are also identical. U6/H7 repeat the uniform/diverse comparison with 5% of each monovalent sign, retaining the same mean bare pair amplitude. Equation (19) does not fix actual potential energy, local solvation, effective excluded volume, or friction: the distribution of interactions still changes. H5/H7 are deliberately strong model perturbations, while M5 is a moderate counterpart.

B. Simulation and Entropy Integration

LAMMPS simulations [12] use $N = 4000$, including 120 probes of each divalent sign. Each composition has three independent replicas, with seeds 64123, 76237, and 91817. Within each replica index, compositions share initial particle positions and probe identities, enabling paired comparisons. Species assignments are random on a perturbed cubic lattice, followed by energy minimization. All particles are integrated with a single Nosé–Hoover NVT thermostat, damping time 0.1τ, and timestep 0.002τ.

Each replica covers 19 densities:

$$\begin{aligned}\rho^* = {} & 0.001, 0.0025, 0.005, 0.01, 0.02, 0.05,\\ & 0.10, 0.15, 0.20, 0.25, 0.30, 0.35,\\ & 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70.\end{aligned} \tag{20}$$

Every state uses the same LJ potential, cut and shifted at 2.5σ, with no tail correction. PPPM accuracy is 10^{-5}; the real-space Coulomb cutoff is $5(0.7/\rho^*)^{1/3}\sigma$, below half the box length. Only the numerical electrostatic splitting changes with density. It must be distinguished from a change in the LJ Hamiltonian, which invalidated the original density-integration path. The Coulomb energy prefactor is one, giving $\lambda_B/\sigma = 0.5$ at $T^* = 2$.

The 18 lower-density stages use 50 000 equilibration and 100 000 production steps each. At $\rho^* = 0.7$, 100 000 equilibration steps precede 500 000 production steps (1000τ); this transport run supplies the final thermodynamic point. All 24 allocations and 456 state stages completed and passed output verification.

TABLE I. Selected replicated MD systems. Every mixture contains $x_{+2} = x_{-2} = 0.03$ and has unit masses and nominal diameters. The primary neutral solvent and all ions have $\varepsilon = 1$. Fractions of secondary neutral solvents and their respective ε values are listed in matching order; $x_{\pm 1}$ is the fraction of each monovalent sign. Label count and mechanically distinct species count are separated. $I = \frac{1}{2}\sum_i x_i q_i^2$.

State	Labels/distinct	$x_{\varepsilon=1,\,q=0}$	Secondary fractions	Secondary ε	$x_{\pm 1}$	I
B3	3/3	0.94	—	—	0	0.12
B5	5/5	0.84	—	—	0.05	0.17
U4	4/4	0.64	0.30	0.25	0	0.12
L5	5/4	0.64	0.15+0.15	0.25, 0.25	0	0.12
M5	5/5	0.64	0.15+0.15	0.0625, 0.5625	0	0.12
H5	5/5	0.64	0.15+0.15	0.01, 0.81	0	0.12
U6	6/6	0.54	0.30	0.25	0.05	0.17
H7	7/7	0.54	0.15+0.15	0.01, 0.81	0.05	0.17

Entropy integration uses the final half of production at each density. The default all-particle temperature has $3N - 3$ kinetic degrees of freedom after removal of global momentum. In reduced units, the configurational pressure reference is

$$Z_{\mathrm{ref}} = 1 + \frac{\langle P\rangle - \rho(1 - 1/N)\langle T\rangle}{\rho T_0}, \qquad T_0 = 2. \tag{21}$$

Subtracting the sampled kinetic contribution and restoring the analytic ideal term removes its finite-N offset and sampling noise from the excess EOS. It does not remove finite-size effects in configurational correlations.

For $a_{\mathrm{ex}} = A_{\mathrm{ex}}/(Nk_BT_0)$, define $y = \sqrt{\rho^*}$ and integrate

$$a_{\mathrm{ex}}(\rho^*) = \int_0^{\sqrt{\rho^*}} \frac{2[Z_{\mathrm{ref}}(y^2) - 1]}{y}\, dy. \tag{22}$$

The Debye–Hückel endpoint is finite in this variable:

$$F(0) = -\frac{[4\pi\lambda_B^* \sum_i x_i q_i^2]^{3/2}}{12\pi}. \tag{23}$$

A cubic spline through this endpoint and the simulated densities gives a_{ex}, followed by $s_{\mathrm{ex}}/k_B = \langle U_{\mathrm{ex}}\rangle/(Nk_BT_0) - a_{\mathrm{ex}}$. Shape-preserving interpolation and alternative production-averaging windows test sensitivity. The ideal reference is always the ideal mixture at the same composition; its mixing entropy is excluded from s_{ex}.

C. Transport Estimators and Replica Uncertainty

Unwrapped particle positions are recorded every 0.5τ for the full dense production run. The same whole-system center-of-mass displacement is subtracted from every particle. Multiple time origins, approximately 200 per lag, provide species MSDs out to 250τ. A linear slope divided by six over 20–100τ is primary; 10–50τ and 50–200τ are sensitivity windows. At every lag and for the fitted coefficients, $D_{\mathrm{net}} = \sum_i x_i D_i$ is checked. Since both probe populations are equal, $D_{\mathrm{probe}} = (D_{+2} + D_{-2})/2$; this is a mean self-diffusion coefficient, not diffusion of a neutral bound pair.

Velocities sampled every 0.01τ during the final 100τ permit species VACF/Green–Kubo checks, using the recorded whole-system COM velocity. All ionic velocities and up to 200 representative particles of each neutral species are retained. Integrals at 5 and 10τ are diagnostics; the short velocity record does not establish a precise long-time plateau. Low-wavevector concentration fluctuations are monitored as a homogeneity diagnostic.

Uncertainty comes from independent replicas, not overlapping MSD origins. Absolute-state intervals use Student-t statistics across three replicas. Composition effects use paired entropy differences and paired $\ln(D_a/D_b)$, with nominal 95% intervals and two degrees of freedom. Percent changes and interval endpoints are obtained by exponentiation. No multiple-comparison correction is applied; finite-size and estimator bias are outside these intervals. L5 is an exact mechanical duplicate of U4 and is never counted as another independent state in scaling fits.

IV. RESULTS AND DISCUSSION OF THE REPLICATED STUDY

A. Higher Excess Entropy Selected by HNC

Table II reports the new dense-state results; these replace the earlier single-trajectory transport estimates as the evidence for component-dependent mobility. The corresponding absolute-state intervals are given in Table III. HNC successfully selects higher-excess-entropy liquids: H5 relative to U4 has $\Delta s_{\mathrm{ex}} = +0.1107k_B$ in HNC and $+0.10322k_B$ in MD. With monovalent background, H7 relative to U6 gives $+0.1108k_B$ and $+0.10331k_B$, respectively. Absolute HNC entropies remain more negative than MD by approximately 0.064–$0.093k_B$ per particle. Agreement of increments is better than agreement of absolute values.

Relative to their original baselines, H5 and H7 raise MD excess entropy by $0.30214k_B$ and $0.30149k_B$. The

TABLE II. Dense-state HNC excess entropy and replicated MD means at $T^* = 2$, $\rho^* = 0.7$, $N = 4000$. Diffusion coefficients are in σ^2/τ; $D_{\mathrm{probe}} = (D_{+2} + D_{-2})/2$. MD uses the primary 20–100τ MSD fitting window. L5 is an exact mechanical duplicate of U4 and is excluded from entropy-scaling fits. Confidence intervals for the primary comparisons are reported in Table IV; absolute-state intervals are in Table III.

State	HNC s_{ex}/k_B	MD s_{ex}/k_B	D_{net}	D_{probe}	D_{+2}	D_{-2}
B3	-2.06350	-1.97101	0.22253	0.20677	0.21196	0.20158
B5	-2.06770	-1.97421	0.22279	0.21047	0.20879	0.21214
U4	-1.84340	-1.77209	0.25945	0.23684	0.23913	0.23455
L5	-1.84340	-1.77166	0.25945	0.23684	0.23913	0.23455
M5	-1.81020	-1.74170	0.26720	0.24038	0.23937	0.24139
H5	-1.73270	-1.66887	0.28401	0.24825	0.24764	0.24885
U6	-1.84780	-1.77603	0.26018	0.23572	0.23742	0.23401
H7	-1.73700	-1.67272	0.28196	0.24795	0.24634	0.24956

TABLE III. Absolute-state nominal 95% Student-t intervals across three independent replicas. Because the states are paired by replica, interval overlap here is not a test of the paired differences in Table IV. These intervals exclude systematic uncertainty.

State	s_{ex}/k_B	D_{net}	D_{probe}
B3	$[-1.97339, -1.96864]$	$[0.21959, 0.22546]$	$[0.20364, 0.20991]$
B5	$[-1.97657, -1.97186]$	$[0.22095, 0.22463]$	$[0.20626, 0.21468]$
U4	$[-1.77342, -1.77076]$	$[0.25884, 0.26005]$	$[0.23336, 0.24032]$
L5	$[-1.77421, -1.76912]$	$[0.25884, 0.26005]$	$[0.23336, 0.24032]$
M5	$[-1.74422, -1.73917]$	$[0.26358, 0.27082]$	$[0.22969, 0.25107]$
H5	$[-1.66936, -1.66838]$	$[0.28214, 0.28588]$	$[0.23921, 0.25728]$
U6	$[-1.77667, -1.77539]$	$[0.25480, 0.26557]$	$[0.21782, 0.25361]$
H7	$[-1.67584, -1.66960]$	$[0.27882, 0.28511]$	$[0.23406, 0.26183]$

moderate split M5 raises it by $0.03039k_B$ relative to U4. Entropy changes induced by diversification are therefore substantially larger than the residual quadrature and averaging-window sensitivities assessed below.

B. Label Count, Ideal Mixing, and Physical Diversification

U4 and L5 provide an exact mechanical null. Splitting the secondary solvent into two identical labels increases formal ideal mixing entropy by $0.30 \ln 2 = 0.20794k_B$ per particle if the labels are treated as distinguishable. Nevertheless, their diffusion coefficients are exactly identical in every replica. The measured excess-entropy difference is only $+0.00042k_B$, with a nominal interval $[-0.00219, 0.00304]k_B$, consistent with thermodynamic sampling. Component labels alone do not supply a mechanical driving variable for diffusion.

M5 and H5 have the same mole fractions and ideal mixing entropy as L5, but different interactions. Thus the L5/M5/H5 entropy differences are interaction-induced excess changes, not additional ideal mixing. Adding monovalent salt to the baseline, B3→B5, increases the ideal mixing contribution by $0.38787k_B$ but slightly *lowers* excess entropy: $\Delta s_{\mathrm{ex}} = -0.00320k_B$, interval $[-0.00543, -0.00097]k_B$. Neither mean diffusion (+0.12%) nor probe diffusion (+1.79%) has a resolved change. More charged components therefore do not by themselves guarantee higher excess entropy or faster probes.

C. Divalent-Ion Enhancement: Large and Controlled Changes

The largest design changes produce clear increases in divalent-ion mobility. H5/B3 gives +20.05%, interval $[14.30, 26.09]$%, while H7/B5 gives +17.79%, interval $[11.17, 24.81]$%. Both signs independently increase at the primary window: +16.8% for +2 and +23.4% for −2 in H5/B3, and +18.0% and +17.6% in H7/B5. All four separate-sign paired nominal intervals exclude zero. These comparisons also alter mean interaction strength and local friction, so entropy is a correlate rather than an isolated causal variable.

The more controlled diversification comparison gives a smaller response. U4→H5 preserves charged composition, ionic strength, masses, nominal diameters, and mean bare pair amplitude. Mean diffusion increases by 9.47%, interval $[8.96, 9.97]$%, whereas probe diffusion increases by 4.81%, interval $[0.60, 9.19]$%. With monovalent background, U6→H7 increases mean diffusion by 8.37%, interval $[7.31, 9.44]$%, and probe diffusion by 5.20%, interval $[-6.14, 17.92]$%. The latter ion effect is unresolved. The moderate split gives a clear mean increase of 2.99%, but its 1.48% probe increase is also unresolved. The ions respond less strongly than the solvent-dominated mean.

TABLE IV. Paired composition changes, numerator relative to denominator. Brackets are nominal 95% Student-t intervals from three independent replica pairs (two degrees of freedom), without multiple-comparison correction. Diffusion changes are $100[\exp(\overline{\ln(D_a/D_b)}) - 1]$ and intervals are transformed in the same way. Entropy differences are per particle in k_B. Intervals quantify replica sampling, excluding estimator and finite-size bias.

Comparison	$\Delta s_{\mathrm{ex}}/k_B$ [95% CI]	ΔD_{net} (%) [95% CI]	$\Delta D_{\mathrm{probe}}$ (%) [95% CI]
L5/U4	+0.00042 [−0.00219, +0.00304]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
M5/U4	+0.03039 [+0.02727, +0.03352]	+2.99 [+1.38, +4.62]	+1.48 [−1.57, +4.64]
H5/U4	+0.10322 [+0.10209, +0.10434]	+9.47 [+8.96, +9.97]	+4.81 [+0.60, +9.19]
H5/B3	+0.30214 [+0.29927, +0.30501]	+27.63 [+25.40, +29.90]	+20.05 [+14.30, +26.09]
H7/U6	+0.10331 [+0.09990, +0.10672]	+8.37 [+7.31, +9.44]	+5.20 [−6.14, +17.92]
H7/B5	+0.30149 [+0.29823, +0.30476]	+26.56 [+25.99, +27.13]	+17.79 [+11.17, +24.81]
B5/B3	−0.00320 [−0.00543, −0.00097]	+0.12 [−1.79, +2.06]	+1.79 [−1.37, +5.04]

D. Rosenfeld-Like Correlation and Its Limits

Figure 1 shows a positive entropy–mobility trend for both the mean and divalent probes. All masses, temperatures and densities are equal, so $D_i^R = D_i\rho^{1/3}/\sqrt{T}$ differs from D_i by the same constant; diffusion ratios are unchanged. Descriptive unweighted fits of composition means to

$$\ln D_i^R = \ln A_i + B_i s_{\mathrm{ex}}/k_B \tag{24}$$

give $B_{\mathrm{net}} = 0.80$ and $B_{\mathrm{probe}} = 0.62$ for B3/U4/M5/H5, and 0.78 and 0.55 for B5/U6/H7. All four fits have $R^2 > 0.99$. These fits contain only four or three composition means, exclude the duplicate L5 state, and do not incorporate uncertainty in both coordinates; the high R^2 is descriptive rather than an independent validation of a universal law.

The controlled local response differs from the broad-family fit. For H5/U4, $\Delta \ln D/\Delta(s_{\mathrm{ex}}/k_B)$ is approximately 0.88 for the mean and 0.46 for probes; H7/U6 gives approximately 0.78 and 0.49. The probe response is weaker than the mean response under the same intervention. Mean self-diffusion assigns only 6% weight to divalent particles and cannot determine their mobility separately. The new results support a restricted entropy–mobility association for both observables, while neither component count nor a single bulk entropy establishes a universal species law. No association-lifetime mechanism has yet been measured in this study.

E. Estimator Sensitivity and Charge-Sign Symmetry

The controlled ion point estimates remain positive, approximately 4–6%, across the MSD windows (Table V and Fig. 2). However, H5/U4 loses nominal significance over 50–200τ: +3.94%, interval $[-8.44, 18.00]$%. H7/U6 is unresolved in all three windows. Accordingly the approximately 5% ion effect is a provisional trend, not a converged precise enhancement. VACF integrals support the broad mobility ordering, but do not independently resolve such a small effect with the present final-100τ records.

Charge-sign symmetry remains a convergence diagnostic. The B3 model is invariant under global charge reversal, yet its primary-window estimate gives $D_{+2} - D_{-2} = 0.01039$, nominal replica interval $[0.00291, 0.01786]$. The other states have sign-difference intervals containing zero. Generated inputs use identical masses and LJ parameters for both probe signs, random probe assignments, one all-particle thermostat, and a common COM frame; no sign-dependent input parameter was found. The B3 asymmetry nevertheless flags remaining sampling or uncertainty-estimation limitations. Pooling the signs does not remove that concern. It also does not explain away the large baseline contrasts, for which each sign separately increases.

Across all 24 replicas, switching from cubic to shape-preserving interpolation changes dense-state entropy by at most $0.00032k_B$. Retaining all or 75% of production instead of the final half changes it by at most $0.00165k_B$. These sensitivities are small relative to the $0.103k_B$ diversification increments, but they do not bound finite-size bias or error below the lowest simulated density. Low-wavevector species concentration-fluctuation diagnostics remain of order unity, approximately 0.48–1.80 for individual replica/species values, with no large enhancement in the selected mixtures. One box size and these trajectory lengths cannot exclude slower segregation or establish phase stability.

F. Rehabilitation and Limits of the 2PT Excess Entropy

We reanalyzed the existing final-100τ velocity records of all 24 replicas using a corrected classical two-phase thermodynamic (2PT) estimator. The revised spectral construction follows Sun *et al.* [13], with the equal-partial-volume mixture approximation discussed by Lai *et al.* [14]. Each species is assigned $\bar{V}_i = 1/\rho$; equal nominal diameters do not establish equality of actual partial volumes in these charged, dispersion-diverse liquids. The primary gas/solid partition exponent is $\delta = 1$; the published LJ value $\delta = 1.5$ is an unfitted sensitivity. Neither partition, reference volume, nor entropy offset is adjusted to match thermodynamic integration (TI).

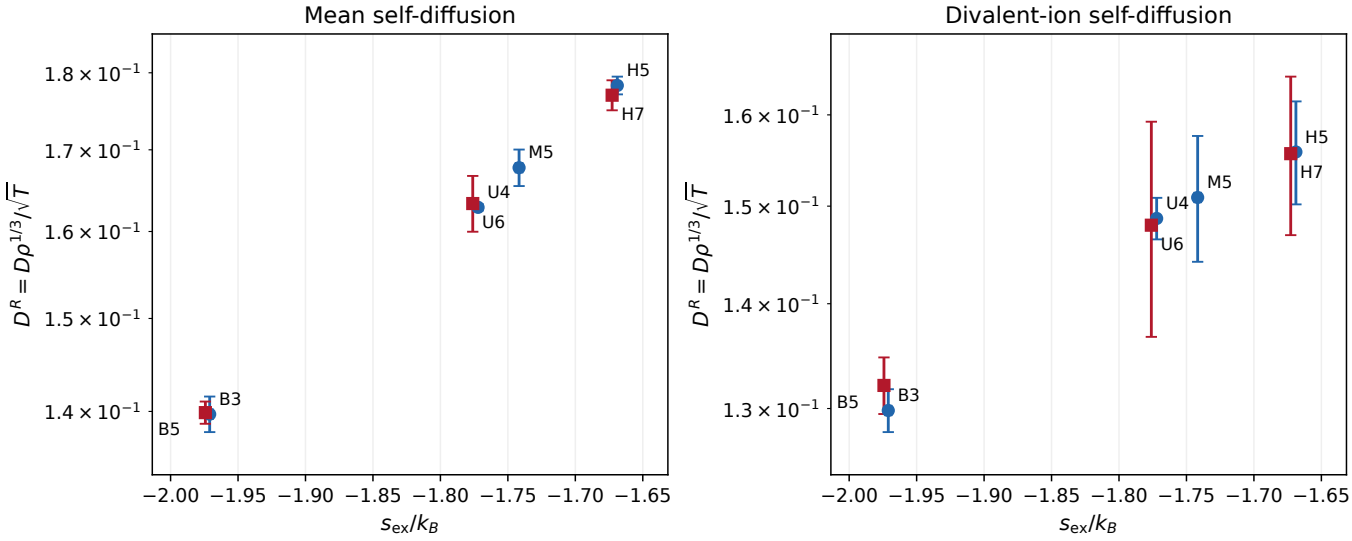


FIG. 1. Rosenfeld-reduced mean and divalent-probe self-diffusion versus replicated MD excess entropy. Vertical axes are logarithmic. Blue circles: no monovalent background; red squares: 5% of each monovalent sign. Bars are nominal 95% absolute-state replica intervals. L5 coincides exactly with U4 in transport and is omitted as a duplicate. The broad positive trend does not fix the slope of controlled composition changes.

TABLE V. Sensitivity of the paired divalent-ion diffusion change (percent) to MSD fitting window, with nominal 95% replica intervals. The 20–100τ window is primary. The positive controlled effect is not uniformly significant across estimators/windows.

Comparison	10–50τ	20–100τ	50–200τ
H5/U4	+5.92 [+1.87, +10.13]	+4.81 [+0.60, +9.19]	+3.94 [−8.44, +18.00]
H7/U6	+5.31 [−1.17, +12.22]	+5.20 [−6.14, +17.92]	+4.46 [−5.66, +15.67]

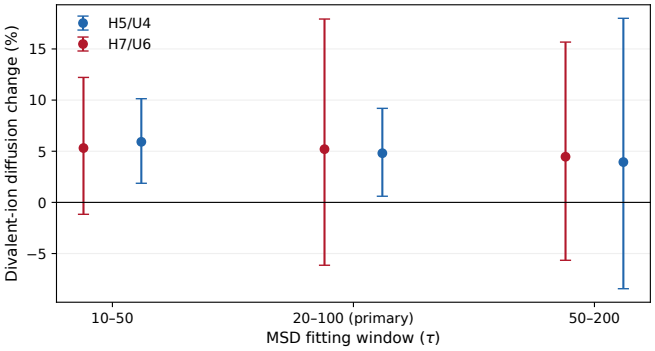


FIG. 2. Controlled divalent-ion diffusion changes and nominal paired 95% intervals for alternative MSD windows. The point estimates are consistently positive, but statistical significance is sensitive to the fitting window and monovalent background.

The calculation corrects the density scaling and prefactor of the normalized diffusivity ($\Delta \propto D\sqrt{m/T}\rho^{1/3}$), weights gas entropy by its gas-like fraction, and subtracts the classical ideal-mixture reference consistently. The complete finite-record autocorrelation, in the common system-COM frame, is Gaussian-tapered with primary width 10τ. Its spectrum spans the full Nyquist band and is normalized by measured $C(0)$ to three modes per particle; raw target-temperature mode counts are retained as a diagnostic. The gas Lorentzian and its entropy tail are integrated analytically to infinity. The residual spectrum is kept signed as revised-2PT bookkeeping, rather than clipped to a positive "solid population." This partition is not a literal count of free ions or bound oscillators.

Seven analytic checks pass, including the standard fluidicity polynomial, unit invariance, a solvable VACF/Green–Kubo limit, the Lorentzian entropy tail, the dilute-gas limit, and classical-reference and label/mixing cancellation. Normalization, reconstruction, and independence of the arbitrary classical phase-space constant hold to floating-point precision. Mechanically identical labels are pooled before the nonlinear entropy estimate. For all three actual U4/L5 pairs, matched retained particle IDs have identical COM-corrected velocity hashes, VACFs, and resulting 2PT entropies. Separately processing the label samples gives a difference $+0.00062k_B$, interval $[-0.00428, 0.00553]k_B$, consistent with sampling. State-mean pooled-probe spectral diffusivities agree with the full-production MSD coefficients within 2.6%.

These successes rehabilitate the implementation, but not its quantitative thermodynamic accuracy across compositions (Fig. 3). For B3 the primary estimate is $-2.07445k_B$, compared with TI $-1.97101k_B$: the approximately $3.4k_B$ discrepancy of the archived $-5.376k_B$ result

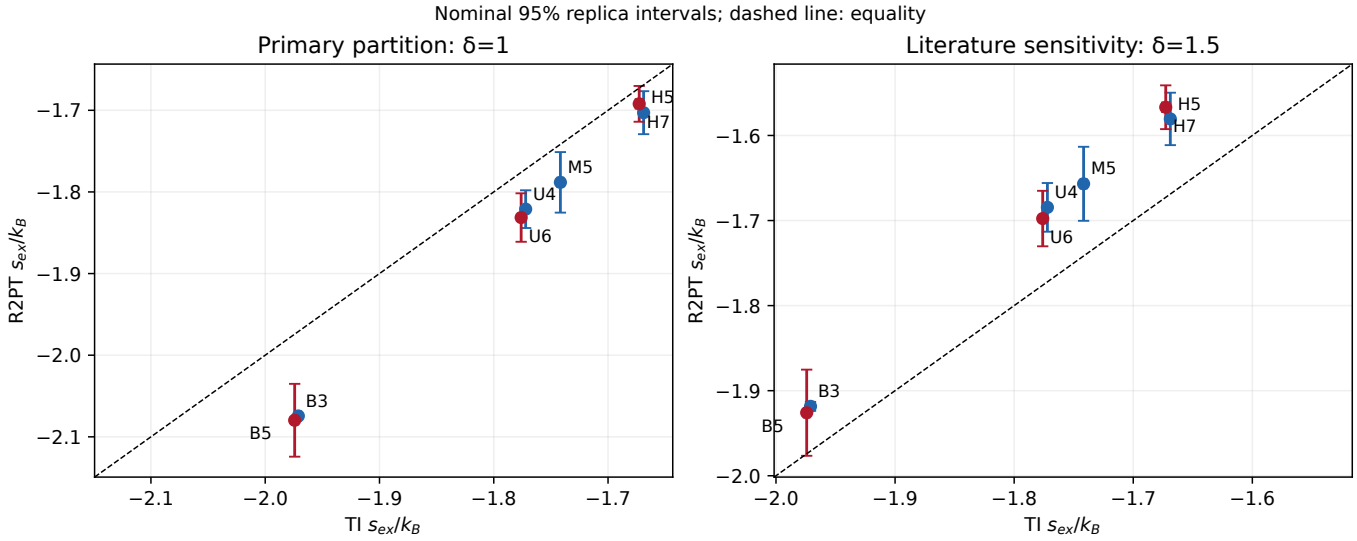


FIG. 3. Corrected classical revised-2PT excess entropy against TI for the seven mechanically distinct mixtures. L5 duplicates U4 in transport and is omitted. Bars are nominal 95% absolute-state replica intervals; dashed lines denote equality. The left panel uses the primary partition $\delta = 1$, and the right panel the unfitted literature sensitivity $\delta = 1.5$. Internal consistency does not remove composition-dependent thermodynamic bias.

is reduced to $0.10344k_B$. Primary absolute biases range from about -0.019 to $-0.105k_B$, with all nominal paired bias intervals below zero. B3, B5, and U6 exceed the prespecified practical mean-bias tolerance $0.05k_B$. Thus the remaining error is composition dependent and comparable to the controlled entropy signal; a constant offset is insufficient.

For H5/U4, primary 2PT gives $\Delta s_{\mathrm{ex}} = 0.11818k_B$ versus TI $0.10322k_B$, a point bias $+0.01496k_B$ (Table VI). This meets the practical point-bias tolerance $0.02k_B$ and is statistically consistent with zero bias, but the wide interval does not establish equivalence within that tolerance. For H7/U6, the estimate is $0.13927k_B$ versus $0.10331k_B$, an approximately 35% overestimate; its bias interval $[0.01501, 0.05692]k_B$ excludes zero. The $\delta = 1.5$ partition improves the first increment to $0.10397k_B$ but still overestimates the second at $0.13086k_B$. Agreement for one pair does not validate a partition for all compositions.

The discrepancies survive sensitivity checks. State-mean entropy spans over taper widths 5, 10, and 20τ are 0.011–$0.022k_B$; first/last-half record shifts reach $0.012k_B$. Frequency-grid refinement changes means by less than $0.00002k_B$, and sampling-interval decimation changes them by at most $0.0022k_B$. Illustrative volume-conserving $\pm 10\%$ partial-volume perturbations shift means by at most $0.0032k_B$, without establishing that the equal-volume approximation is exact. Positive-part clipping would add modes: the signed negative residual represents approximately 2.3–3.9% of the total mode weight in primary species spectra. The corrected bookkeeping avoids this artifact, but cannot eliminate the model dependence of the gas/solid partition.

We therefore retain TI as the entropy coordinate in the Rosenfeld analysis. Corrected 2PT and the underlying VACFs remain useful dynamical diagnostics, but species spectral allocations are not independently measured partial molar entropies. Because diffusion enters the zero-frequency spectrum and fluidicity construction, a 2PT–diffusion correlation is not an independent confirmation of entropy-driven enhancement. Probe fluidicity rises from about 0.530 in B3 to 0.569 in H5, consistent with increased low-frequency motion. However, the mean probe VACFs remain positive over the first τ; a pronounced early negative backscattering minimum is not supported. Association populations and residence times, rather than a literal interpretation of the 2PT partition, are needed to test local-caging mechanisms.

G. Further Discriminating Checks

The first priority is longer independent matched replicas of U4/H5 and U6/H7, together with extended B3 sampling to resolve the charge-sign flag. Next, repeat matched states with $N = 8000$ or 16000 at fixed density and composition. Box-size effects need not cancel between liquids with different viscosity. Particle-size variation is a separate perturbation of packing, contact electrostatic coupling, and friction and should not be conflated with this box-size test.

Existing trajectories permit probe–solvent and probe–ion RDFs, coordination populations, association lifetimes, and exchange-time analysis. Species-resolved pair entropy could help explain local mobility, but is not a thermodynamic partial molar entropy or a replacement for total excess entropy. These measurements would test local caging and association mechanisms without inferring ki-

TABLE VI. Controlled excess-entropy increments from TI and corrected classical revised 2PT. All entropy quantities are per particle in k_B. Bias is the paired difference between the 2PT and TI increments; brackets are nominal 95% Student-t intervals from three independent replica pairs (two degrees of freedom), without multiplicity correction. The primary partition is $\delta = 1$; $\delta = 1.5$ is an unfitted sensitivity.

Comparison	Partition	TI $\Delta s_{\mathrm{ex}}/k_B$	2PT $\Delta s_{\mathrm{ex}}/k_B$	Bias [95% CI]
H5/U4	$\delta = 1$	0.10322	0.11818	$+0.01496$ $[-0.00544, +0.03536]$
H5/U4	$\delta = 1.5$	0.10322	0.10397	$+0.00075$ $[-0.02175, +0.02326]$
H7/U6	$\delta = 1$	0.10331	0.13927	$+0.03596$ $[+0.01501, +0.05692]$
H7/U6	$\delta = 1.5$	0.10331	0.13086	$+0.02755$ $[+0.00705, +0.04805]$

netics from static entropy alone.

After convergence tests, increasing $x_{+2} = x_{-2}$ symmetrically, for example to 0.06 or 0.10, would strengthen the ionic contribution and alter screening and association. Uniform/diverse solvents should be compared at matched concentration and ionic strength within each pair, with HNC re-screening and homogeneity checks. Several densities and temperatures at fixed composition, and states with matched excess entropy but different compositions, would distinguish state-point scaling from composition-induced correlations.

V. CONCLUSIONS

The replicated calculations establish a positive excess-entropy–mobility association in the selected solvent-design families, with clear limits on its interpretation.

1. **Component count alone has no demonstrated transport effect.** An exact label-only split raises formal ideal mixing entropy while leaving diffusion identical. Monovalent salt addition slightly lowers excess entropy and gives no resolved divalent speedup.

2. **HNC-guided interaction changes produce higher-excess-entropy, faster-ion liquids.** MD confirms entropy increases of approximately $0.30k_B$ and divalent diffusion gains of 18–20% relative to the original baselines. Both signs separately increase. These are successful design comparisons, not isolated entropy interventions.

3. **Controlled diversification gives a weaker, provisional ion response.** At fixed charged composition and mean bare pair amplitude, excess entropy rises by approximately $0.103k_B$, mean diffusion by 8–9%, and probe diffusion by approximately 5%. The ion significance is sensitive to fitting window and unresolved with monovalent background. Longer independent sampling and the B3 charge-sign convergence check remain necessary.

4. **Rosenfeld-like trends extend to probes but do not establish a universal law.** Broad-family fits have positive slopes and high descriptive R^2, with weaker slopes for probes than for the mean. Three or four composition means cannot determine universality, and changing interactions also changes local friction and solvation. Species-resolved kinetics, box-size comparisons, and state-point validation are the next discriminating tests.

5. **Corrected 2PT is internally consistent but not a quantitative TI replacement.** All 24 records pass analytic/reference and mechanical-label checks, while state-mean probe spectral diffusion agrees with MSD within 2.6%. Entropy biases nevertheless vary with composition: the controlled monovalent-background increment is overestimated by approximately 35% in the primary model. The partition sensitivity does not consistently remove this error. We retain TI for the entropy–mobility comparison and use spectra as supporting dynamics, not an independent scaling validation or a literal free-ion fraction.

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Appendix A: Historical Cutoff Audit and Superseded Estimates

The earlier $N = 2000$ comparison comprised a 94% neutral-solvent 3C baseline, two 4C liquids containing 89% primary solvent and 5% secondary solvent with $\varepsilon = 1.25$ or 0.75, and the 84%-solvent 5C baseline with 5% of each monovalent sign. All contained 3% of each divalent sign. These 4C states differ from U4 and the new 30%-secondary-solvent designs.

The original density sweep changed the LJ cutoff with density, so integrating $(Z-1)/\rho$ did not follow a single Hamiltonian. Repeating all 60 states with a common shifted LJ cutoff 2.5σ corrected that error. Retaining the original pressure estimator and finite-intercept quadrature isolates cutoff-only dense-state entropy shifts of -0.08717, -0.08565, -0.08501, and $-0.09107k_B$ for 3C, strongly

attractive 4C, weakly attractive 4C, and 5C. The dense-state interactions and energies were already unchanged; the correction arose from lower-density pressure integration. Approximately half of the former HNC–MD entropy gap was removed.

With the historical target-temperature mean-pressure estimator and screened integration, the corrected $N = 2000$ entropies were -1.96693, -1.97261, -1.96199, and $-1.97588 k_B$. These values belong to the historical protocol: they do not use the new configurational pressure reference, particle number, density grid, or replica sampling. The new B3 and B5 values in Table II supersede them for the present comparisons. The 4C values remain an auxiliary dispersion comparison and are not combined with the new transport estimates.

The archived species MSDs used separate group-COM subtraction and were not consistent with all separately reported VACF/2PT transport values. Those estimates and unverified species-entropy decompositions have therefore been removed from the quantitative transport argument. No physical charge-asymmetric caging mechanism or universal Rosenfeld exponent is inferred from them. The common-frame replicated trajectories now provide the primary evidence; their own remaining convergence limitations are reported explicitly.

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